New interactions of high-energy cosmic neutrinos, from the Hubble scale to the Earth

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(Dated: November 7, 2026)

[(1) The question: light mediators of new neutrino interactions act on high-energy astrophysical neutrinos all along their trip. (2) What we do: follow the flux from the sources through the Universe, the Milky Way, the Sun and the Earth, with the potential and the collisions on the relic neutrinos together, for three families of models. (3) What we fit: IceCube's flavor composition and spectrum, and its starting and through-going events. (4) What we find: which stage of the trip matters where, and the resulting limits, with where they compete. (5) Context: future telescopes.]

I. INTRODUCTION

Do neutrinos feel new interactions, carried by light particles that couple to them? Many extensions of the Standard Model contain such a mediator: a gauge boson of a difference between lepton numbers [1–3], a dark gauge boson that reaches the neutrinos through their mixing with a dark neutrino [4, 5], or a scalar or vector through which neutrinos interact among themselves [6]. Because the mediator is light, its effects reach far. It can act over long distances, or resonate when neutrinos collide with other neutrinos at specific energies. Either way, its effects build up over the distance that the neutrinos travel.

Laboratory experiments, neutrino oscillations, and cosmology constrain these mediators strongly over wide ranges of mass and coupling [7–11]. These probes work over terrestrial or solar distances, or in the early Universe, where their reach depends on assumptions about its history. Effects that accumulate over cosmological distances at high energies lie beyond their reach. So does a test of these interactions that is independent of the early Universe.

High-energy astrophysical neutrinos offer such a test [12, 13]. Discovered by IceCube [14], they reach the Earth with energies from TeV to PeV, after traveling up to gigaparsecs. IceCube now measures both their energy spectrum and their flavor composition [15–17]. On their way, a new mediator can act on them through the potential of the matter that they cross, which changes their flavor, and through collisions with the relic neutrinos left over from the early Universe [18, 19].

Earlier studies treated these effects one at a time and simplified the trip. Studies of the potential evaluated it only at the detector [20]. Studies of the collisions took the relic neutrinos at rest [21, 22], until a recent one let the lightest of them move [23].

[Fig. 1 (money plot): how each stage of the trip moves the limits; $L_e - L_\mu$, IceCube HESE + DR2; full calculation, the potential at the detector only, and one intermediate step; existing bounds in gray.]

FIG. 1. [Headline; what the curves are; see Sec. V; takeaway.]

In this work, we follow high-energy neutrinos along their whole trip, through the Universe, the Milky Way, the Sun, and the Earth, computing the potential and the collisions together. We apply this calculation to the three families of mediators above and fit it to several IceCube data sets: the flavor composition and the spectrum, and the starting and through-going events. **How a new mediator changes the flux at Earth depends on where along the trip it acts, so reliable limits need the full trip.** [Figure 1 shows . . . : a paragraph that describes it in full, once the money-plot runs are done.]

Figure 2 shows the trip, as an artist's rendition in four panels that zoom in from one to the next. In the Universe, the neutrinos travel from their sources through the relic neutrinos, with which they can collide, and through the potential of the Universe's electrons. In the Milky Way, they cross the potential of its stars and gas, joined by the neutrinos that the Galaxy emits. Near the Sun, the potentials of the Sun, the Moon, and the Earth switch on. Inside the Earth, the neutrinos cross its layers on

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their way to IceCube, at the South Pole. Of the mediators above, only the gauge boson of a difference between lepton numbers couples to ordinary matter, which sources these potentials; every mediator acts through collisions with the relic neutrinos.

Section II introduces the mediators and the processes through which they act. Section III describes the trip, Sec. IV the data and the statistical method, and Sec. V our results. Section VI summarizes. The appendices give the vertices of each model, the transport in full, the flux at the stops of the trip, the thermal rates, the decays in a medium, the transport for secret interactions, and our treatment of the Earth.

II. NEW LIGHT MEDIATORS

In each model that we study, a light particle couples to neutrinos and mediates a new interaction. We consider three families of such mediators, which act on high-energy neutrinos in different ways. In the first, a new gauge boson couples to the difference between electron number and muon or tau number [1–3]. This boson couples to electrons as well as to neutrinos, so the electrons in ordinary matter source a potential that changes the flavor of the neutrinos. In the second, the dark portal, a dark gauge boson reaches the known neutrinos only through their mixing with a new, dark neutrino [4, 5]. In the third, secret interactions, a scalar or vector mediator couples to neutrinos alone [6]. For the latter two, ordinary matter plays no role; only the relic neutrinos act on the high-energy neutrinos, by colliding with them.

Figure 3 shows every process that our calculation includes, each drawn once with a vector mediator and labeled with the models that it belongs to: (a) the potential of ordinary matter; (b) resonant absorption on a relic neutrino; (c, d) scattering on a relic, through the t and u channels; (e) the production of a pair of mediators; (f) the decay of the mediator; (g) the decay of a neutrino into a lighter one and the mediator; and (h) the longitudinal mode of a vector mediator. Of these, only the potential requires ordinary matter; the rest require only the relic neutrinos. Appendix A redraws the processes for a scalar.

$L_e - L_\alpha$.—Under the symmetry $L_e - L_\alpha$, with $\alpha = \mu$ or τ, the new boson Z', of mass M, couples with strength g' to the electron and its neutrino, and with the opposite sign to the charged lepton ℓ_α and its neutrino:

$$\mathcal{L}_{L_e-L_\alpha} = -g' Z'_\mu \big(\bar{e}\gamma^\mu e + \bar{\nu}_e \gamma^\mu P_L \nu_e - \bar{\ell}_\alpha \gamma^\mu \ell_\alpha - \bar{\nu}_\alpha \gamma^\mu P_L \nu_\alpha\big) \,, \tag{1}$$

where P_L projects onto left-handed states. A neutrino at position $\mathbf{r}$ feels the potential of the electrons within a distance of about $1/M$, the range of the interaction [Fig. 3(a)] [20]:

$$\mathbf{V}(\mathbf{r}) = g'^2 \, \Phi_e(\mathbf{r}) \, \mathbf{S}_\alpha \,, \qquad \Phi_e(\mathbf{r}) = \int d^3 r' \, n_e(\mathbf{r}') \, \frac{e^{-M|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r}-\mathbf{r}'|} \,, \tag{2}$$

where n_e is the number density of electrons, $\mathbf{S}_\mu = \mathrm{diag}(1,-1,0)$ and $\mathbf{S}_\tau = \mathrm{diag}(1,0,-1)$ in the flavor basis. Antineutrinos feel $-\mathbf{V}$. The electrons of the Universe, of the Milky Way, of the Sun, of the Moon, and of the Earth all contribute. Each body contributes once the range reaches it, so the lighter the mediator, the more distant the matter that the neutrinos feel. Above twice the electron mass, the mediator also decays into charged-lepton pairs [Fig. 3(f)].

The dark portal.—In the dark portal, no Standard Model particle carries charge under the new gauge symmetry, $U(1)_D$. A dark neutrino carries it and mixes with the known neutrinos. The dark gauge boson, Z_D, has mass M and coupling g_D. It reaches the known neutrinos through the mixing of the dark neutrino with each flavor α, $U_{\alpha 4}$ [4, 5]:

$$\mathcal{L}_{\mathrm{DP}} = -Z_{D\mu} \sum_{\alpha\beta} G_{\alpha\beta} \, \bar{\nu}_\alpha \gamma^\mu P_L \nu_\beta \,, \qquad G_{\alpha\beta} = g_D \, U_{\alpha 4} U^*_{\beta 4} \,. \tag{3}$$

The coupling matrix G has rank one. We let the dark neutrino mix with one flavor at a time, so that the interaction has strength $g_D |U_{\alpha 4}|^2$ on ν_α. In the model of Ref. [5], Z_D also mixes with the Z boson, which would let matter source a potential, as for $L_e - L_\alpha$. That mixing is at most $(v_\varphi/v_H)(M/m_Z)$, which follows from Eq. (6) of Ref. [5] and the structure of its gauge sector. Here, v_φ and v_H are the vacuum values of the Higgs doublets with and without dark charge, and m_Z is the mass of the Z boson. Their ratio, v_φ/v_H, is at most a few. At the light masses where a potential could act, this bound lies 8 to 30 orders of magnitude below the mixing that IceCube data could detect. We set the mixing to zero, so only the relic neutrinos act on the high-energy neutrinos.

Secret interactions.—Here a new mediator couples to neutrinos alone, with a flavor structure set by a matrix G. A scalar, ϕ, and a vector, V, couple as

$$\mathcal{L}_\phi = -\kappa \, \phi \sum_{\alpha\beta} G_{\alpha\beta} \, \bar{\nu}_\alpha \nu_\beta \,, \qquad \mathcal{L}_V = -V_\mu \sum_{\alpha\beta} G_{\alpha\beta} \, \bar{\nu}_\alpha \gamma^\mu P_L \nu_\beta \,, \tag{4}$$

with $\kappa = 1/2$ for Majorana neutrinos and $\kappa = 1$ for Dirac neutrinos. We take G equal to g times $\mathrm{diag}(1,0,0)$, $\mathrm{diag}(0,1,0)$, $\mathrm{diag}(0,0,1)$, or the identity, so that the mediator couples with strength g to one flavor or to all of them equally. With two mediators and two kinds of neutrino, these choices give 16 models. For Dirac neutrinos, the scalar flips chirality: each vertex joins a left-handed

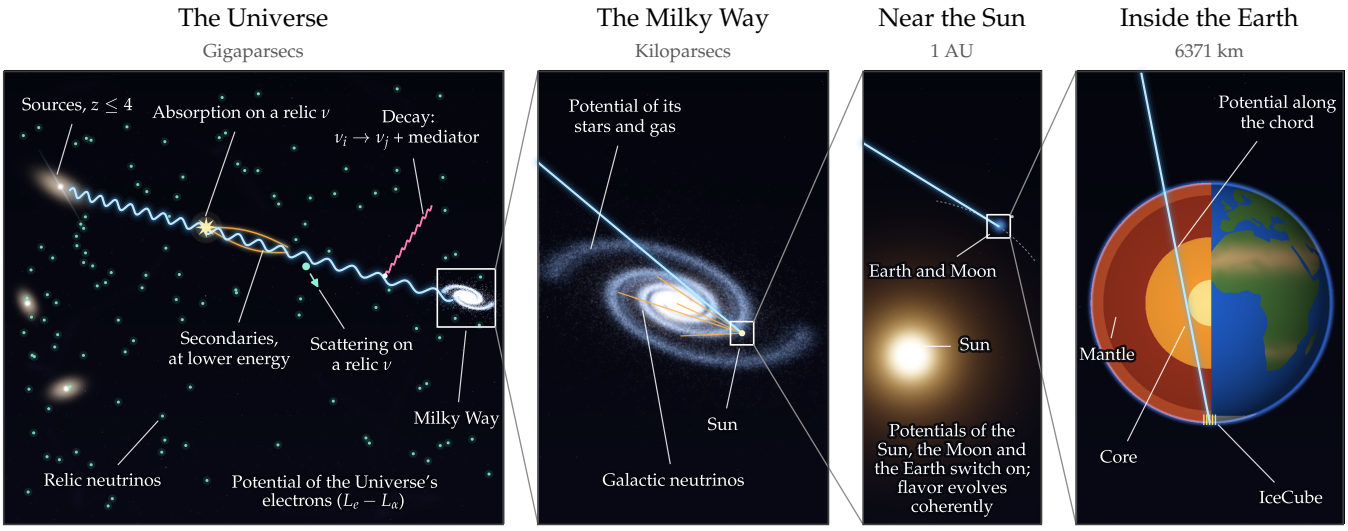


FIG. 2. ***The trip of high-energy astrophysical neutrinos, and what happens along it.*** An artist's rendition, not to scale; each panel zooms in on the next. In the Universe, neutrinos from sources out to $z = 4$ can be absorbed on relic neutrinos, which returns secondaries at lower energy, scatter on them, or decay into a lighter neutrino and the mediator, while the potential of the Universe's electrons acts on them for $L_e - L_\alpha$. In the Milky Way, its stars and gas add their potential; the Galaxy also adds its own neutrinos. From the Sun on, the potentials of the Sun, the Moon, and the Earth change the flavor coherently, along each path through the Earth to IceCube. Which of these processes act depends on the model and on the mass of the mediator (Sec. II). See Sec. III for details.

neutrino, which interacts weakly, to a right-handed one, which does not. Collisions then turn part of the high-energy flux into right-handed neutrinos, which IceCube cannot detect. Dirac relic neutrinos also keep the helicity they had when they decoupled, so moving relics present the two helicities in unequal proportions (Appendix D).

What each mediator does, by mass.—The mass of the mediator decides which of the processes in Fig. 3 matter. A high-energy neutrino of energy E meets a relic neutrino of mass m_r at resonance when $E = M^2/(2m_r)$. For mediator masses of roughly a keV to tens of MeV, this resonance falls inside IceCube's energy range for at least one relic mass state, depending on the neutrino masses. For instance, for $M = 10$ MeV, the resonances on the two heavier relics fall at 1.0 and 5.8 PeV. There, absorption carves dips in the spectrum [Fig. 3(b)], while the decays of the mediator return neutrinos of lower energy [Fig. 3(f)]. For lighter mediators, the high-energy neutrinos also scatter on the relics with small momentum transfer, which moves them between states [Fig. 3(c, d)]. If the mediator is lighter than the difference between two neutrino masses, the heavier neutrino can decay into the lighter one and the mediator [Fig. 3(g)]. For vector mediators, the rates of these processes grow as the mediator becomes lighter, through its longitudinal mode.

Where the models hold.—When the neutrino masses break the symmetry behind the current to which a vector couples, the longitudinal mode of the vector couples to neutrinos like a scalar at energies far above M [11]. For the vectors of Eqs. (1), (3), and (4), its couplings to the neutrino mass states ν_i and ν_j are [24]

$$Y_{ij} = \frac{i}{M}\left(G'_{ij}\, m_i + G'_{ji}\, m_j\right) \,, \tag{5}$$

where m_i is the mass of ν_i and G' is the coupling matrix in the basis of mass states [Fig. 3(h)]. These couplings grow as $1/M$. Tree-level perturbative unitarity requires the largest partial-wave amplitude that they produce to stay below 1/2 in magnitude [24]. For $L_e - L_\mu$, this excludes $g' > 10^{1.76}\, M/\text{eV}$ when the lightest neutrino is nearly massless; heavier neutrinos lower this line. We mark the excluded region in our figures and draw no conclusions from it. The scalar has no longitudinal mode, so it has no such region.

III. THE TRIP

What we follow.—We follow neutrinos and antineutrinos separately, from their sources to the detector (Fig. 2). Over cosmological distances, the phases of their oscillations average out [25], so the number of neutrinos in each of three states describes the flux fully. These states are the eigenstates $|a\rangle$, with $a = 1, 2, 3$, of the Hamiltonian

$$H(E, z) = \frac{U\,\mathrm{diag}(m_1^2, m_2^2, m_3^2)\,U^\dagger}{2E(1+z)} + \mathbf{V}(z) \,, \tag{6}$$

where U is the lepton mixing matrix, E is the energy observed today, z is the redshift, and $\mathbf{V}$ is the potential. For antineutrinos, $U \to U^*$ and $\mathbf{V} \to -\mathbf{V}^*$. Without a

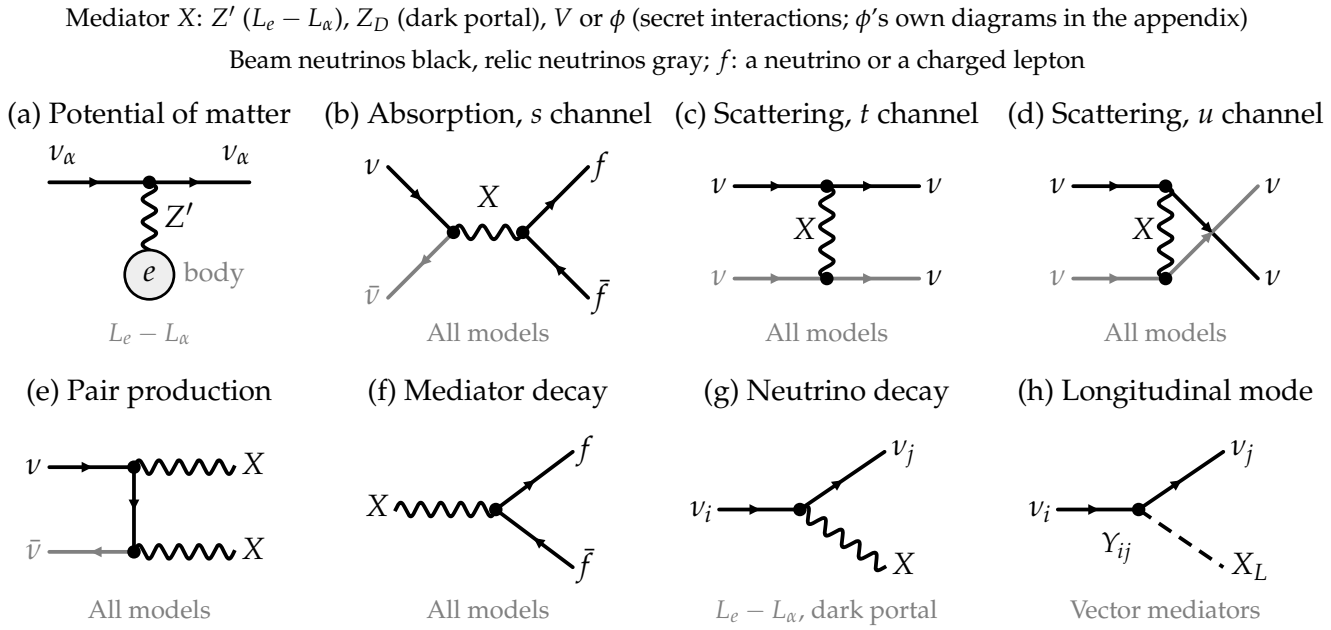


FIG. 3. ***The processes of the trip.*** Every process that our calculation includes, with the models it belongs to under each diagram. The mediator is drawn as a vector; for a scalar, the same diagrams hold with a dashed line, and Fig. 10 shows them for Majorana neutrinos. Pair production, (e), proceeds also through the diagram with the two mediators exchanged. Only the potential, (a), needs ordinary matter; every other process needs only the relic neutrinos. See Sec. II for details.

potential, the eigenstates are the mass states; where the potential dominates, they are nearly pure flavor states. In normal ordering, ν_1, ν_2, and ν_3 become ν_μ, ν_τ, and ν_e for $L_e - L_\mu$, and ν_τ, ν_μ, and ν_e for $L_e - L_\tau$. For antineutrinos, the potential has the opposite sign, which exchanges the roles of ν_1 and ν_3. The cosmological potential varies on the scale of the Hubble time, far more slowly than the neutrinos oscillate, so each neutrino stays in its eigenstate as the potential evolves. Only collisions and decays move neutrinos between states. We follow the number of neutrinos in each state until they reach the Sun, where we switch to a coherent description.

Sources.—We place the sources up to $z = 4$, with a number density that tracks the star-formation rate [26]. Each emits neutrinos and antineutrinos with a power-law energy spectrum and a flavor composition $(f_{e,S}, f_{\mu,S}, f_{\tau,S})$. At the source, this mixture of flavors splits incoherently over the eigenstates there, each flavor ν_α contributing to state a in proportion to $|\langle\nu_\alpha|a\rangle|^2$. In the analysis (Sec. IV), the spectrum is a broken power law, built from solutions at fixed spectral index.

The cosmological transport.—From the sources to today, the number $F_a(E, z)$ of neutrinos per unit energy in state a evolves as

$$\frac{\partial F_a}{\partial t} = \mathcal{S}_a - \Gamma_a^{\rm abs} F_a + \mathcal{R}_a + \sum_{b\neq a} (R_{ab}F_b - R_{ba}F_a) + \frac{\partial}{\partial E}(b_a F_a) - \Gamma_a^{\rm dec} F_a + \mathcal{D}_a \,, \tag{7}$$

with $dt = -dz/[(1+z)H(z)]$ and $H(z)$ the Hubble rate. The terms are, in order: the sources; absorption on the relics [Fig. 3(b)] and the neutrinos it returns, $\mathcal{R}_a$ [Fig. 3(f)]; scatterings on the relics that move neutrinos between states [Fig. 3(c, d)]; the energy that neutrinos lose to relics that change mass in those scatterings; and the decay of state a into lighter ones, together with the daughters $\mathcal{D}_a$ that decays feed into it [Fig. 3(g)]. Because E is the energy observed today, the redshift enters through it, with no separate term. Antineutrinos obey the same equation. The neutrinos returned by absorption couple the two kinds. At MeV masses, where the potential is negligible, the eigenstates are the mass states. There we solve the same equation with every channel of Fig. 3(b)–(e), pair production included (Appendix F). Appendix B gives every term in full.

Absorption on the relics.—State a is absorbed at the rate

$$\Gamma_a^{\rm abs}(E, z) = \sum_r \Gamma_r(E, z)\, \frac{1}{2} \left|(U^\dagger G W)_{ra}\right|^2 \,, \tag{8}$$

where r runs over the relic mass states, W holds the eigenvectors of Eq. (6) in the flavor basis, and Γ_r is the rate of resonant absorption on relic r per unit coupling. Thus, the absorption amplitude combines coherently the flavor content of state a and that of relic r. For instance, in vacuum, a coupling equal for all flavors absorbs mass state j only on relic j. We average Γ_r over the thermal momenta of the relics, at temperature $T_{\nu,0}(1+z)$, with $T_{\nu,0} = 1.68 \times 10^{-4}$ eV today. This average matters most when the lightest relic is relativistic, which broadens its

resonance [23]. At MeV masses, the thermal motion of the relics enters every collision term, not only the resonance. For Dirac relics, we also account for their helicity (Appendix D). Each absorption of a neutrino of energy E' returns a neutrino and an antineutrino, with energy spectra $3E^2/E'^3$ and $3(E'-E)^2/E'^3$.

Decays and scatterings.—For light mediators, two more processes move neutrinos between states. A state can decay into a lighter one by emitting the mediator [Fig. 3(g)]; for Majorana neutrinos, the daughter can also be an antineutrino. Scatterings on the relics that transfer little momentum move neutrinos between states without changing their number [Fig. 3(c, d)]. When the relic changes its mass state in such a scattering, the neutrino also loses energy. In a potential, the states that decay are the eigenstates of Eq. (6), not the mass states (Appendix E). The two processes compound: scatterings refill the states that decay, so together they drain the flux at a given energy more than either does alone.

The Milky Way.—Today, the extragalactic flux enters the Milky Way, whose stars and gas [27] add their own potential to the cosmological one. This potential also changes slowly along the way, so each state passes smoothly into the matching eigenstate of the potential near the Sun. Where the potential of the Milky Way opens decays that the cosmological one does not, we instead follow the flux across the Galaxy direction by direction, with the rates of Eq. (7). The Milky Way also emits neutrinos of its own, made by cosmic rays that collide with its gas and by unresolved sources [28, 29]. We follow them from where they are emitted, along each line of sight, through the same processes. Absorption on the relics is negligible inside the Galaxy, whose column of relics is about 10^{-5} of the cosmological one.

The Sun, the Moon, and the Earth.—From the Sun on, we follow the flux coherently. Its density matrix starts as $\rho = \sum_a N_a |a\rangle\langle a|$, with N_a the number of neutrinos, extragalactic and Galactic, in state a, and evolves as

$$\frac{d\rho}{dx} = -i\,[H(x), \rho] \ , \tag{9}$$

where x is the distance along the path and $H(x)$ is the Hamiltonian of Eq. (6) today, plus the potential of the Standard Model inside the Earth. At the detector, the flux of flavor α is $\langle\nu_\alpha|\rho|\nu_\alpha\rangle$, averaged over the paths through the Earth from each direction. Along the way, the potentials of the Sun, of the Moon, and of the Earth itself switch on, each within its range. Where a potential switches on slowly compared to the oscillations, it turns each state into its own eigenstate. For instance, for $L_e - L_\mu$ at 10^{-20} eV, the potential of the Sun turns the mass state ν_2 into ν_τ. We compute the potential of the Earth from Eq. (2), with the electron density of the Preliminary Reference Earth Model [30]. We neglect collisions and decays from the Sun on; decays would matter there only where the potential already dominates the change of flavor (Appendix E).

The flux along the trip.—Figure 4 shows the flux of each of the six species, the three flavors of neutrinos and of antineutrinos, position by position along the whole trip, for one benchmark of each family: a 10 MeV scalar coupled to ν_τ, for secret interactions; a 10^{-2} eV mediator mixed with ν_τ, for the dark portal; and $L_e - L_\mu$ at 10^{-26} and 10^{-20} eV. Its four columns match the panels of Fig. 2: the Universe, versus redshift; a line of sight across the Milky Way; a path past the Sun; and a chord through the Earth. The sources sit in a shell at $z = 1.8$–2.2. We show one observed energy, 495 TeV, and normalize each flux to the total after the sources stop emitting. Secret interactions and the dark portal, which have no potential, act only in the Universe. For $L_e - L_\mu$, the place where the flavor changes depends on the mediator mass: at 10^{-26} eV, the Milky Way, whose potential also opens decays between the lightest neutrino and antineutrino states that drain part of the flux; at 10^{-20} eV, the neighborhood of the Sun. Appendix C shows how this depends on energy.

The collision terms, one at a time.—Figure 5 shows what each term of Eq. (7) does to the flux at the detector, versus energy, switching the terms on one at a time. Its top row shows the flux of all flavors over the flux with no new interaction; the rows below show the share of each flavor. For a 10 MeV scalar coupled to ν_τ, with $g = 10^{-3}$, the lines add in turn resonant absorption, the other collisions with the relics, and the thermal motion of the relics. For $L_e - L_\mu$ at 10^{-2} eV, with $g' = 10^{-6}$, they add the potential, the decays, and the scatterings on the relics. On the left, absorption opens a dip at each resonance with a ν_2 or ν_3 relic; the neutrinos that it gives back lift the flux just below. Adding the scatterings deepens the loss that the decays cause.

IV. DATA AND STATISTICAL METHOD

Data.—We fit three data sets, each on its own. The first combines two measurements of the diffuse flux by IceCube. One is the flavor composition measured with 11.4 years of medium-energy starting events [17]. We use its likelihood as a function of (f_e, f_μ) and compare it with the flavor composition of our flux between 10 TeV and 30 PeV. The other is the energy spectrum measured with tracks and cascades together, given as the normalization of an E^{-2} flux in separate energy segments [16]. We compute the same normalizations from our flux and compare them with the ten measured ones between 4.6 TeV and 10 PeV, with the asymmetric errors of each segment. The second data set uses events instead: 7.5 years of high-energy starting events [31], whose topologies carry information on flavor, and the through-going muon tracks above 3 TeV of the IceTracks-DR2 release, from its IC79 and IC86 configurations [32]. We fold our flux with the public detector responses of both. This set leaves out the flavor measurement, whose events overlap the starting events. The third data set is a forecast: the events that

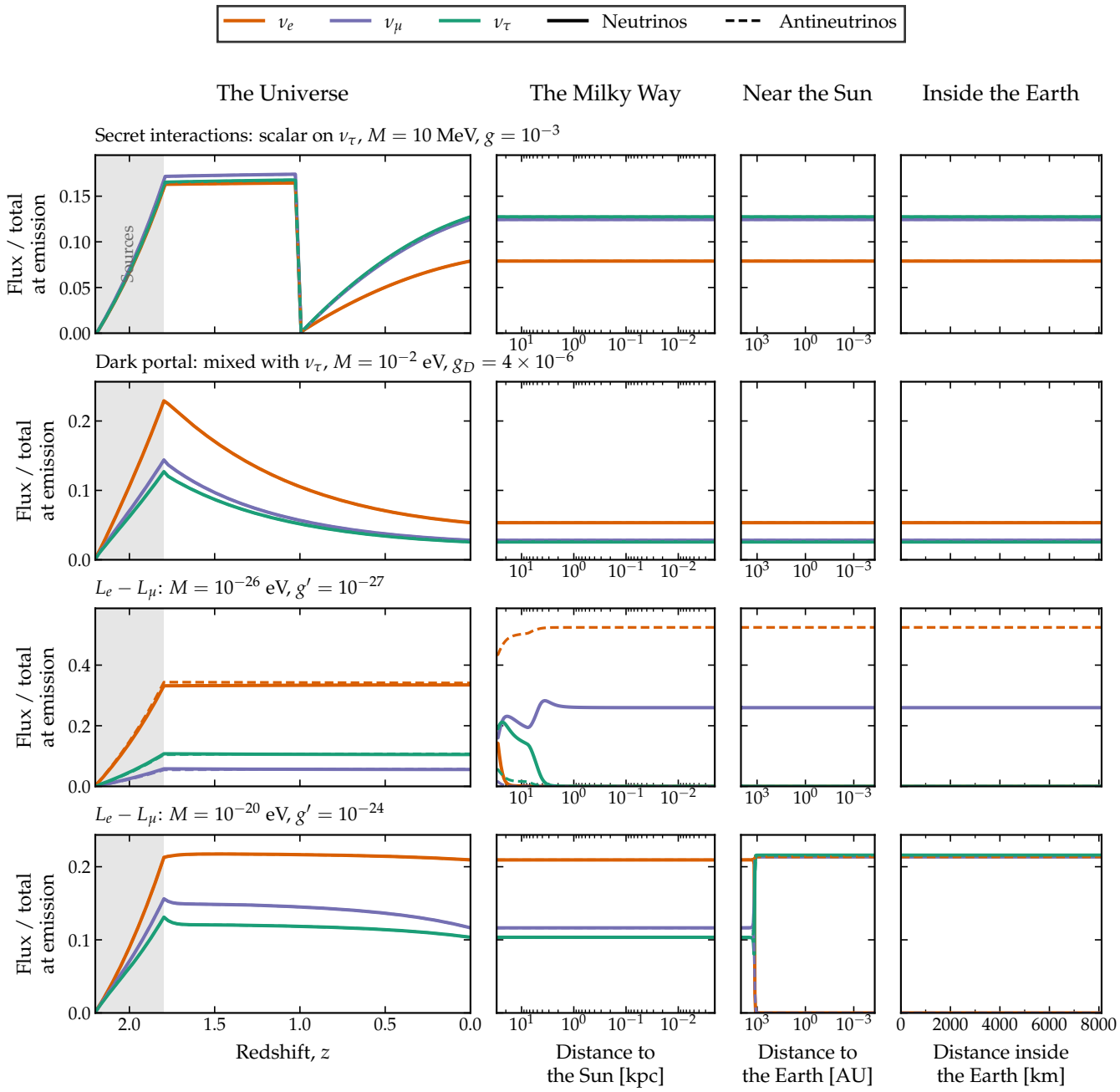


FIG. 4. ***Where along the trip each family of mediators acts.*** The flux of each species, neutrinos and antineutrinos of each flavor, at an observed energy of 495 TeV, from sources in a shell at $z = 1.8$–2.2 (gray) to the detector, divided by the total once the sources stop emitting. The columns follow the panels of Fig. 2: the Universe; one line of sight across the Milky Way; one path past the Sun, perpendicular to the direction to it; and one chord through the Earth, at $\cos\theta_z = -0.64$. Secret interactions absorb the neutrinos resonantly in the Universe, and the secondaries of the absorption partly refill the flux. The dark portal drains it across the Universe through decays and scatterings on the relic neutrinos. For $L_e - L_\mu$, the potential of the Milky Way opens decays that turn neutrinos into antineutrinos at 10^{-26} eV; that of the Sun turns the states into flavor states at 10^{-20} eV. Inside the Earth, nothing changes at these energies.

eight neutrino telescopes would detect over 10 years each, computed without a new interaction: IceCube, IceCube-Gen2 [33], KM3NeT/ARCA [34], Baikal-GVD [35], P-ONE [36], TRIDENT [37], NEON [38], and HUNT [39].

Expected events.—In bin k of an event sample, we ex-

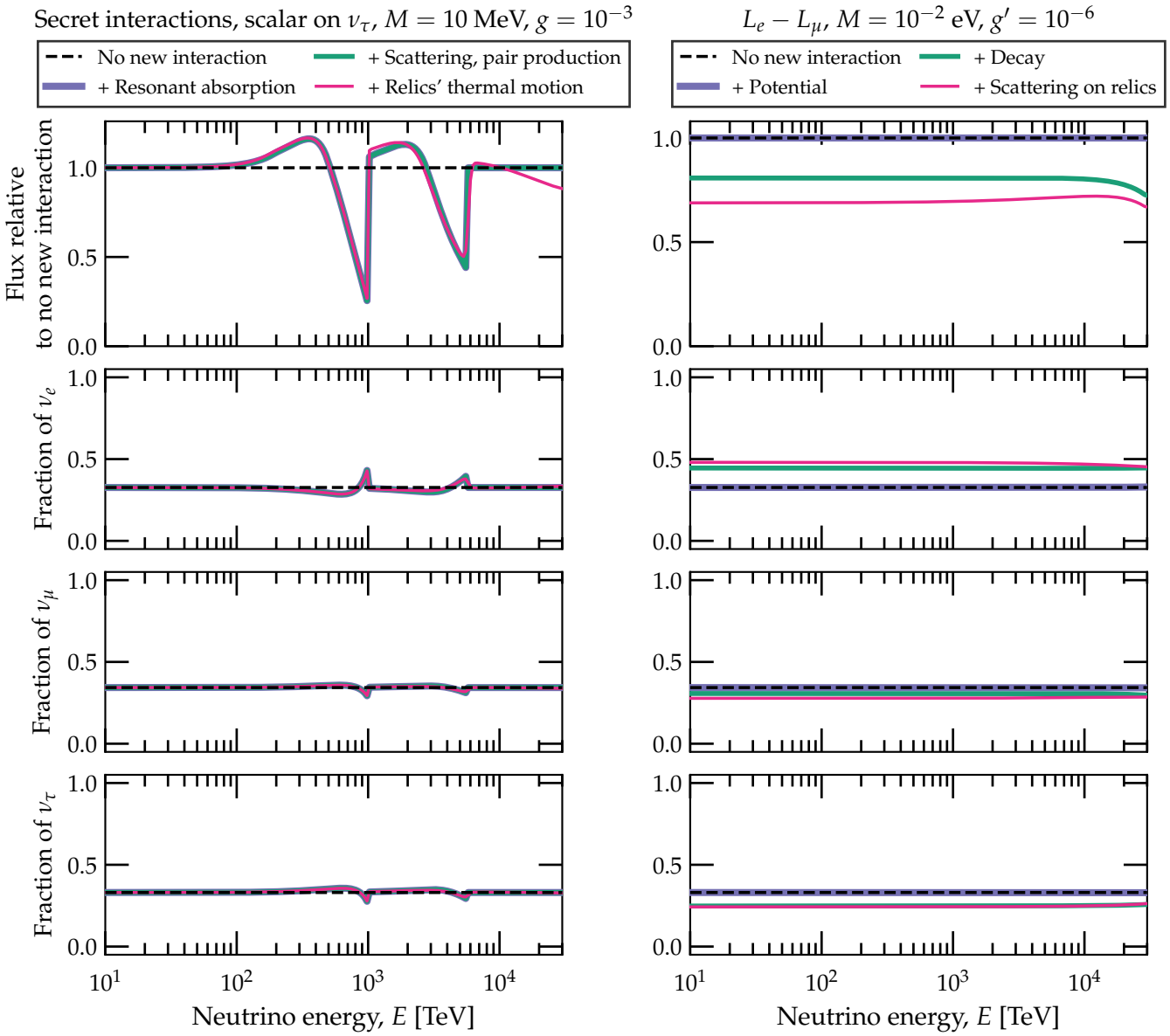


FIG. 5. ***What each term of the transport does to the flux and its flavors.*** *Top:* the all-flavor flux, relative to that without a new interaction. *Below:* the share of each flavor. Each line adds one term to those of the line before. We assume pion-decay sources, $\nu_e : \nu_\mu : \nu_\tau = 1 : 2 : 0$, with an $E^{-2.5}$ spectrum, the normal mass ordering, and the lightest neutrino at 10^{-6} eV. *Left:* absorption carves dips where the neutrinos resonate with the ν_2 and ν_3 relics, while the neutrinos it returns raise the flux below each dip. *Right:* scatterings on the relics, which alone conserve the number of neutrinos, deepen the loss from decays. See Sec. III for details.

pect

$$\mu_k = T \sum_\alpha \int dE\, d\Omega\; \Phi_\alpha(E,\Omega)\, A_\alpha^{(k)}(E,\Omega) + \mu_k^{\rm bkg}\ , \quad (10)$$

where T is the exposure time, Φ_α is the flux of flavor α at the detector, from direction Ω, $A_\alpha^{(k)}$ is the effective area that leads into bin k, and $\mu_k^{\rm bkg}$ counts the background of atmospheric neutrinos and muons. The likelihood of each sample is Poisson, $-2\ln\mathcal{L} = 2\sum_k[\mu_k - n_k + n_k \ln(n_k/\mu_k)]$, with n_k the observed events.

The flux in the fits.—The astrophysical flux is a broken power law, with a free normalization and free spectral indices above and below a break fixed at 24.5 TeV, where IceCube's fit of tracks and cascades places it [15, 16]. The Galactic component [28] has a normalization with a Gaussian prior centered on that fitted by IceCube, 0.91 times the template, with a width of 0.18 [40]. The source flavor composition is either fixed to that of pion decay, $(1/3, 2/3, 0)$, or fitted, with free fractions of ν_e at the extragalactic and at the Galactic sources and no ν_τ. The

lightest neutrino mass varies from 10^{-6} to 0.1 eV, in normal ordering, with a prior from cosmology on the sum of the masses, below 0.12 eV at 95% C.L. [41]. We fix the mixing parameters to the best fit of NuFIT 6.1 [42]. [The size of the effect of the mixing parameters on the limits, estimated separately.]

Backgrounds and detector systematics.—For the event samples, the background of atmospheric neutrinos carries the uncertainties of the hadronic yields in air showers, parametrized as in Ref. [43], together with the tilt of the spectrum of cosmic rays and the normalization of the prompt neutrinos. For the starting events, we also vary the normalizations of the conventional atmospheric neutrinos and of the atmospheric muons; for the tracks, the normalization of each configuration of the detector. We minimize over all of them, each with its prior.

How a limit is read.—At each mediator mass M and coupling g, we minimize $\chi^2 = -2\ln\mathcal{L}$ over every other parameter, and compute

$$\Delta\chi^2(M,g) = \chi^2(M,g) - \min_{g'} \chi^2(M,g') \,. \tag{11}$$

A coupling is disfavored at 95% C.L. when $\Delta\chi^2 \geq 3.84$. For light mediators whose decays change the flux at every coupling in our range, the reference is instead the fit without a new interaction. We do not use the region beyond perturbative unitarity (Sec. II). The threshold of 3.84 follows from Wilks's theorem, which may fail here, since the coupling has a boundary and has no effect below some value. [Coverage tests: pseudo-experiments at chosen true values; the coverage found; whether a Feldman–Cousins construction is needed; what is conservative.]

V. RESULTS

What each stage does to the limits.—[Fig. 1; the earlier standard, in-line formula.]

$L_e - L_\mu$ and $L_e - L_\tau$.—[Figure 6 shows . . . (a paragraph that describes it in full); existing bounds [7, 8, 11, 44–47].]

The dark portal.—[Figure 7 shows . . . (a paragraph that describes it in full); existing bounds [9, 10, 48].]

Secret interactions.—[Figure 8 shows . . . (a paragraph that describes it in full); existing bounds [6, 9, 49]; the relativistic relic [23].]

IceCube's MeV preference, or the discovery case.—[Figure 9 shows . . . (a paragraph that describes it in full).]

Future telescopes.—[Projected regions in Figs. 6–8.]

VI. SUMMARY AND OUTLOOK

[A recap for a fresh reader; main findings in **bold**; limitations; outlook; a closing statement that makes a new point.]

Appendix A: The vertices of each model

The transport works in the basis of neutrino mass states, ν_i, related to the flavor states by $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$. For the vector mediators of Eqs. (1), (3), and (4), the mass states couple through the Hermitian matrix $G' = U^\dagger G U$, with $G = g'\mathbf{S}_\alpha$ for $L_e - L_\alpha$. For a scalar and Majorana neutrinos, they couple through the complex symmetric matrix $\lambda = U^T G U$. This matrix is real only without CP violation. For $\delta_{\rm CP} = 212°$, even a flavor-universal coupling has off-diagonal entries of 9–13% of g. No rate depends on the Majorana phases. When the scalar couples to a single flavor α, so does $\delta_{\rm CP}$, since then $|\lambda_{ij}| = g\,|U_{\alpha i}||U_{\alpha j}|$. For a scalar and Dirac neutrinos, the coupling is $-\phi\sum_{ij} h_{ij}\,\bar\nu_i\nu_j$, with $h = U^\dagger G U$. We give the right-handed neutrinos the same mixing as the left-handed ones, since nothing observable fixes their basis.

We evaluate the diagrams with Majorana neutrinos by the fermion-flow rules of Ref. [50]. For Dirac neutrinos, the scalar changes chirality at each vertex, connecting a state that feels the weak interaction, ν_L or $\bar\nu_R$, with a sterile one, ν_R or $\bar\nu_L$. A relic at rest presents both helicities to the high-energy neutrino. Part of the secondaries of each collision are then sterile; the transport tracks them apart from the visible ones, per mass state and energy.

Figure 10 shows, in its top row, how the mediator of each model couples, as in Eqs. (1), (3), and (4). Its bottom row redraws the processes of Fig. 3 for a scalar, with Majorana neutrinos: each scalar vertex joins two neutrinos, so a high-energy neutrino is absorbed on a relic neutrino, not on a relic antineutrino.

The longitudinal mode of a vector mediator couples through the divergence of the current it couples to. For Majorana neutrinos, written as left-handed Weyl fields χ_i, the current $\sum_{ij} G'_{ij}\,\chi_i^\dagger\bar\sigma^\mu\chi_j$ has divergence $i\sum_{ij} G'_{ij}(m_i\chi_i\chi_j - m_j\chi_i^\dagger\chi_j^\dagger)$. At energies far above M, the longitudinal mode then acts as a scalar with the couplings Y_{ij} of Eq. (5), through the term $-\frac{1}{2}\sum_{ij} Y_{ij}\,\phi\,\chi_i\chi_j + \text{h.c.}$ in the Lagrangian [24]. We require the largest eigenvalue of the tree-level partial-wave matrices of these couplings, for two-fermion states with total angular momentum $J = 0$ and for fermion-scalar states with $J = 1/2$, to stay below 1/2 in magnitude [24].

Appendix B: The transport equation in full

This appendix gives every term of Eq. (7) and the transport inside the Milky Way. The neutrinos and the antineutrinos are followed as two coupled populations, F_a and $\bar F_a$; we write the terms for neutrinos.

Sources.—Sources of comoving number density $R(z)$, distributed as the star-formation rate [26], each emit $\Phi_0 E_s^{-p}$ neutrinos per unit energy E_s at the source. Split

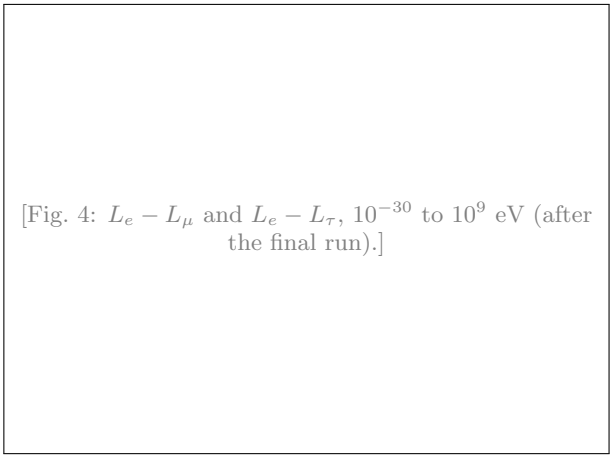


FIG. 6. [Caption.]

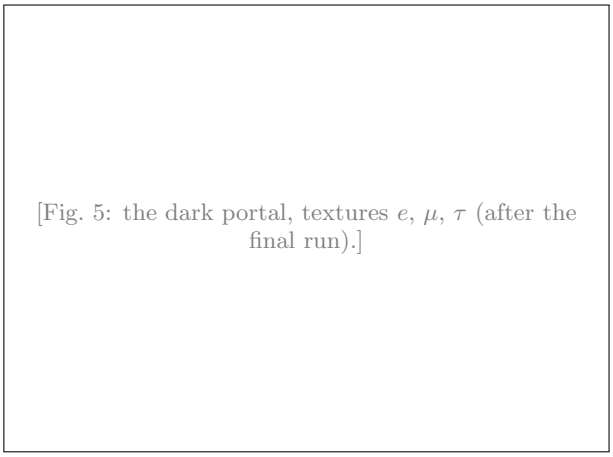


FIG. 7. [Caption.]

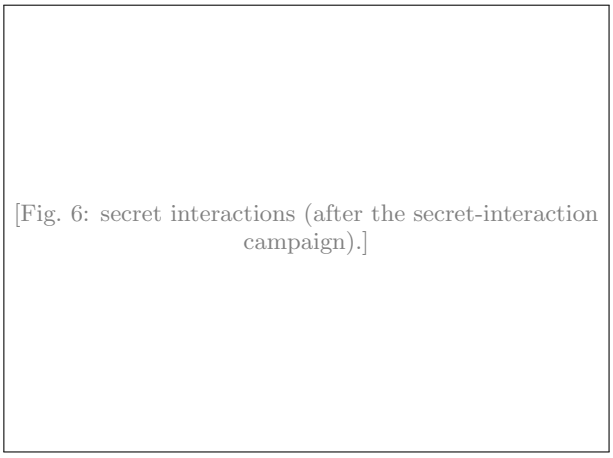


FIG. 8. [Caption.]

over the eigenstates at the source,

$$\mathcal{S}_a(E,z) = w_\nu\,\Phi_0\,[E(1+z)]^{-p}\,R(z)\,(1+z) \times \sum_\alpha f_{\alpha,S}\,|\langle\nu_\alpha|a\rangle|^2\,, \tag{B1}$$

where w_ν and $w_{\bar\nu}$ weigh the two kinds and $(1+z)$ converts the energy at the source into that observed today. The eigenstates are those of the potential at the sources.

Absorption and its secondaries.—The rate of absorp-

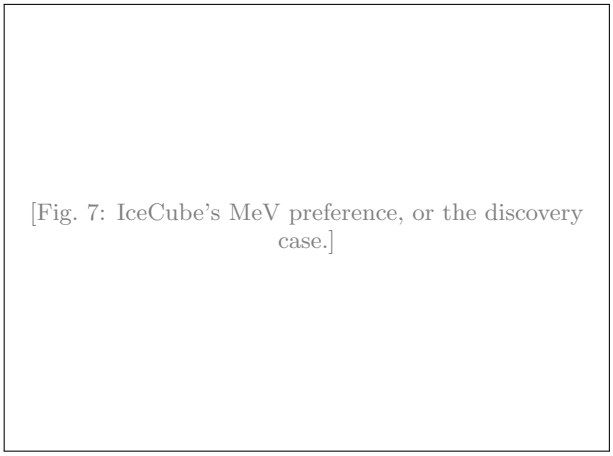


FIG. 9. [Caption.]

tion is that of Eq. (8). An absorption gives back one neutrino and one antineutrino. The neutrinos returned into state a at energy E are

$$\mathcal{R}_a = \beta_a \int_E^\infty dE' \left[\frac{3E^2}{E'^3} \sum_c \Gamma_c^{\rm abs} F_c + \frac{3(E'-E)^2}{E'^3} \sum_c \bar{\Gamma}_c^{\rm abs} \bar{F}_c \right]_{E'} , \tag{B2}$$

where the first term comes from absorbed neutrinos and the second from absorbed antineutrinos. The share of the secondaries that enter state a, $\beta_a = {\rm Re}(W^\dagger G G^\dagger W)_{aa}/[{\rm Tr}(GG^\dagger) + 24\pi\Gamma_{\rm vis}/M]$, follows from the coupling of the mediator to that state, with $\Gamma_{\rm vis}$ its width into charged leptons. Summed over states, it is the branching ratio of the mediator into neutrinos.

Scatterings on the relics.—They move neutrinos from state a to state b at the rate

$$R_{ba} = \Gamma_{\rm u}(z)\, |G_{ba}|^2 \, \frac{\omega_{ab}^2}{\omega_{ab}^2 + D_{ab}^2} , \tag{B3}$$

where $G_{ba} = \langle b|G|a\rangle$, and ω_{ab} is the difference between the energies of the two states. The damping D_{ab} combines the rates at which scatterings and decays remove neutrinos from the two states. When it exceeds the splitting ω_{ab}, the conversions are suppressed. The rate per unit coupling squared, $\Gamma_{\rm u}(z) \propto n_{\rm hel}(1+z)^3/[M^2 + m_D^2(z)]$, grows with today's number density of relics in each mass state and helicity, $n_{\rm hel} = 3\zeta(3) T_{\nu,0}^3/(4\pi^2)$, and falls with the mass of the mediator, increased by its screening mass m_D in the gas of relics.

Energy loss to relics that change mass.—A scattering that leaves the relic in a heavier mass state costs the neutrino energy. Averaged over the relics, this lowers the observed energy at the rate

$$b_a(E,z) = \frac{C(z)\,(GG^\dagger)_{aa}}{1+z} , \tag{B4}$$

where $C(z)$ sums, over every pair of relic mass states $r \neq s$, the rate of such scatterings times the energy they cost, $(m_s^2 - m_r^2)/(2E_r)$, averaged over the thermal energies E_r of the relics.

Decays.—The loss and gain terms of decays are those of Appendix E.

The Milky Way.—Inside the Galaxy, the number $G_a(E,\ell,\hat{n})$ of neutrinos that travel toward the Sun along direction $\hat{n}$, at distance ℓ from it, evolves as

$$-\frac{\partial G_a}{\partial \ell} = j_a + \sum_{b\neq a} (R_{ab} G_b - R_{ba} G_a) + \frac{\partial}{\partial E}\,(b_a G_a) - \Gamma_a^{\rm dec} G_a + \mathcal{D}_a , \tag{B5}$$

with every rate evaluated today, in the eigenstates of the local potential. Absorption is left out: the Galaxy holds a column of relics about 10^{-5} times that of the Universe. The Galaxy emits, per unit length,

$$j_a = \tfrac{3}{2}\, N_{\rm G}\, I(E,\hat{n})\, [w_{\rm gas}\, q_{\rm gas}(\ell,\hat{n}) + (1-w_{\rm gas})\, q_{\rm src}(\ell,\hat{n})] \times \sum_\alpha f_\alpha\, |\langle \nu_\alpha | a\rangle|^2 , \tag{B6}$$

where I is the intensity per flavor, neutrinos plus antineutrinos, of the template of Ref. [28], so that the factor 3/2 gives all three flavors of one kind. The normalization $N_{\rm G}$ is free in the fits; $w_{\rm gas}$ is the share of the emission from the gas. The distributions $q_{\rm gas}$ and $q_{\rm src}$ of the emission from the gas and from unresolved sources along the line of sight are normalized to one; $(f_e, f_\mu, f_\tau) = (x, 1-x, 0)$ is the flavor composition at emission. Where the Galactic potential allows decays that the cosmological potential forbids, the extragalactic flux crosses the Galaxy by the same equation without emission, along every line of sight, and is then averaged within each band of declination.

Solving it.—We solve Eq. (7) on 439 energies, from 4.6 TeV to 30 PeV, and 200 steps in redshift, from $z = 4$ to today. The eigenstates are tabulated on a grid in σ, since they depend on energy and redshift only through it. Near the Earth, we average over 8 paths past the Sun and over 12 chords through the Earth in each band of declination.

Appendix C: The flux at the stops of the trip

Figure 11 shows the flux and its flavor composition at four stops of the trip, from 10 TeV to 30 PeV: as it reaches the Milky Way, after crossing it, near the Earth, and at the detector. Its top row shows the flux of all flavors relative to that on arrival at the Milky Way; the rows below show the fraction of each flavor. Its two columns are the masses of $L_e - L_\mu$ in Fig. 4, with the same couplings. It complements Fig. 4, which follows one energy along the trip: at 10^{-20} eV, the flavor composition changes at the stop near the Earth; at 10^{-26} eV, at the stop after the Milky Way, where the flux also drops.

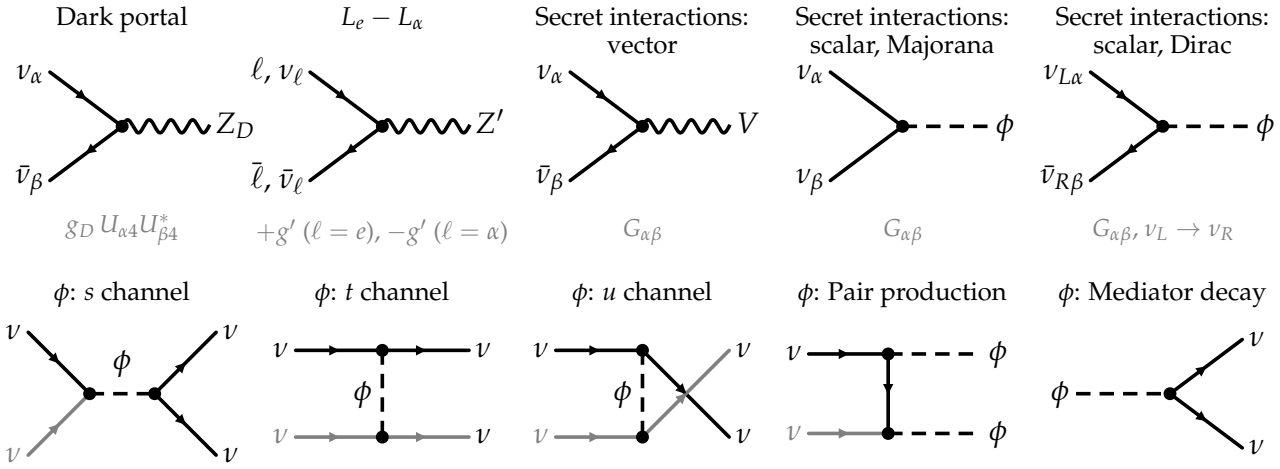


FIG. 10. ***The vertices of each model, and the diagrams of the scalar mediator.*** *Top:* the vertex of each model, with its coupling from Eqs. (1), (3), and (4). *Bottom:* the processes of Fig. 3 with a scalar mediator, for Majorana neutrinos. On this vertex two neutrinos meet, so the absorption starts from a relic neutrino rather than an antineutrino. For Dirac neutrinos, every scalar vertex turns a left-handed neutrino into a right-handed one. See Appendix A for details.

Appendix D: Thermal absorption rates and the relics' helicity

Absorption on a relic of mass m_r is a resonance at $s = M^2$, where s is the square of the energy of the collision in its center-of-momentum frame. For a relic at rest, $s = 2E(1+z)\, m_r$. The relics, however, move, with Fermi–Dirac momenta at the temperature $T_{\nu,0}(1+z)$. A relic of energy E_p and momentum p that meets the high-energy neutrino at an angle θ gives $s = m_r^2 + 2E(1+z)\,\epsilon$, with $\epsilon = E_p - p\cos\theta$. Thus, the motion of the relics spreads the resonance over a range of energies, most of all for a relativistic lightest relic [23]. The resonance is far narrower than any grid in energy, so we integrate the Breit–Wigner rate over the momenta of the relics semi-analytically, treating the narrow zone around the resonance with exact antiderivatives. For a vector coupled to left-handed neutrinos, the width of the mediator is $\Gamma = M\,\mathrm{Tr}(GG^\dagger)/(24\pi)$, plus its width into charged leptons where those decays are open.

At MeV masses, we average every term of the collision over the momenta of the relics, not only the resonance. Each product of a collision is boosted along the incoming neutrino by a Lorentz factor of about $E/\sqrt{s} \sim 10^8$. Its share of the energy is the same in the frame of the relic as in the cosmological frame, so the energy spectra of the products depend on s alone. Averaging over the momenta of the relics then reduces to averaging over s.

A Dirac relic retains its helicity from the time of decoupling: $-1/2$ for a neutrino and $+1/2$ for an antineutrino. At rest and averaged over directions, it is unpolarized; in motion, its spin follows its momentum. The spin of a relic neutrino along the direction of the incoming neutrino is

$$\pi = \frac{E_p\,\epsilon - m_r^2}{p\,\epsilon}\,, \tag{D1}$$

from -1 for a relic that moves along with the incoming neutrino to $+1$ for one that moves against it. After averaging over the azimuth of the final state, the squared amplitude of a collision is

$$|\mathcal{M}|^2 = \frac{1+\pi}{2}\,|\mathcal{M}_L|^2 + \frac{1-\pi}{2}\,|\mathcal{M}_R|^2\,, \tag{D2}$$

up to terms of order m_r^2/s, where $\mathcal{M}_L$ and $\mathcal{M}_R$ are the amplitudes for a massless relic of left- or right-handed chirality at the same s. A vector couples only to the left-handed part; a scalar couples to both, with the visible and sterile secondaries of Appendix A. We take the density of relics everywhere as the cosmological mean, neglecting their gravitational clustering in the Milky Way [51–53].

Figure 12 shows the consequences for a ν_e-coupled mediator, with Dirac neutrinos: the flux over that with no new interaction, versus energy, for a scalar (top) and a vector (bottom), at $M = 31.6$ keV with $g = 2\times10^{-3}$ and $M = 4.22$ MeV with $g = 1.35\times10^{-3}$ (left and right), for lightest neutrino masses of 10^{-6} eV (thick lines) and 0.03 eV (thin lines). Its lines compare relics at rest, relics in motion without their helicity, and relics in motion with it. With a relativistic lightest relic, the helicity matters most: the scalar's resonance on it nearly disappears, while the vector's deepens.

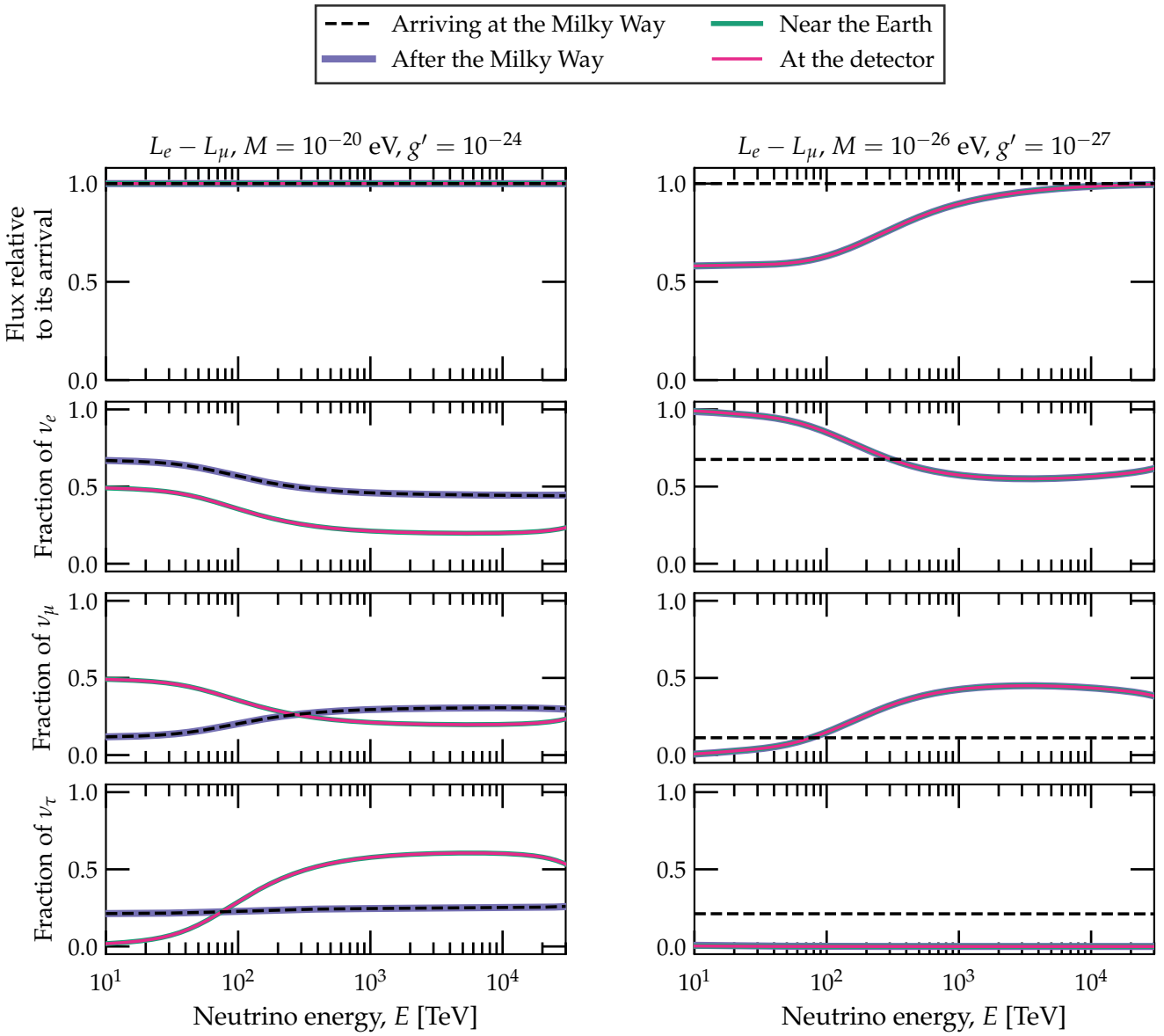


FIG. 11. ***Flux and flavor composition of high-energy neutrinos at the stops of their trip.*** *Top:* the flux of all flavors, neutrinos plus antineutrinos, relative to its value on arrival at the Milky Way. *Below:* the fraction of each flavor. The sources emit $\nu_e : \nu_\mu : \nu_\tau = 1 : 2 : 0$ with a spectrum $E^{-2.5}$; the neutrino masses follow the normal ordering, with a lightest mass of 10^{-6} eV. See Sec. III and Appendix C for details. Which stop shapes the flux depends on the range of the mediator: the Sun, for $M = 10^{-20}$ eV, and the Milky Way, for $M = 10^{-26}$ eV.

Appendix E: Decays in a medium

A neutrino ν_i decays into a lighter ν_j and the mediator when $m_i > m_j + M$ [Fig. 3(g)]. In vacuum, a neutrino of observed energy E' at redshift z decays at the rate $\Gamma_{i\to j} = \Gamma_{ij}^{\text{rest}}\, m_i/[E'(1+z)]$, with $\Gamma_{ij}^{\text{rest}}$ the rate at rest. The daughter takes a fraction $x = E/E'$ of its parent's energy, distributed as

$$\frac{d\Gamma_{i\to j}^{\text{keep, flip}}}{dE} = \frac{\Gamma_{i\to j}}{E'}\left[a + b\,x + \frac{c}{x}\right]_{ij}^{\text{keep, flip}} \qquad \text{(E1)}$$

between the kinematic limits x_- and x_+, with coefficients a, b, c that follow from the helicity amplitudes of the decay. The daughter keeps the helicity of its parent ("keep") or, for Majorana neutrinos, flips it and emerges as an antineutrino ("flip"). Every decay keeps one neutrino, so decays conserve the number of neutrinos while they lower their energy.

Inside a potential, decays connect the eigenstates of Eq. (6), since those are the states in which the neutrinos propagate. Writing the potential as $\mathbf{V} = A\,\mathbf{S}$, the eigenvalues of $\text{diag}(m_i^2) + \sigma\, U^\dagger \mathbf{S} U$, with $\sigma = 2E(1+z)A$,

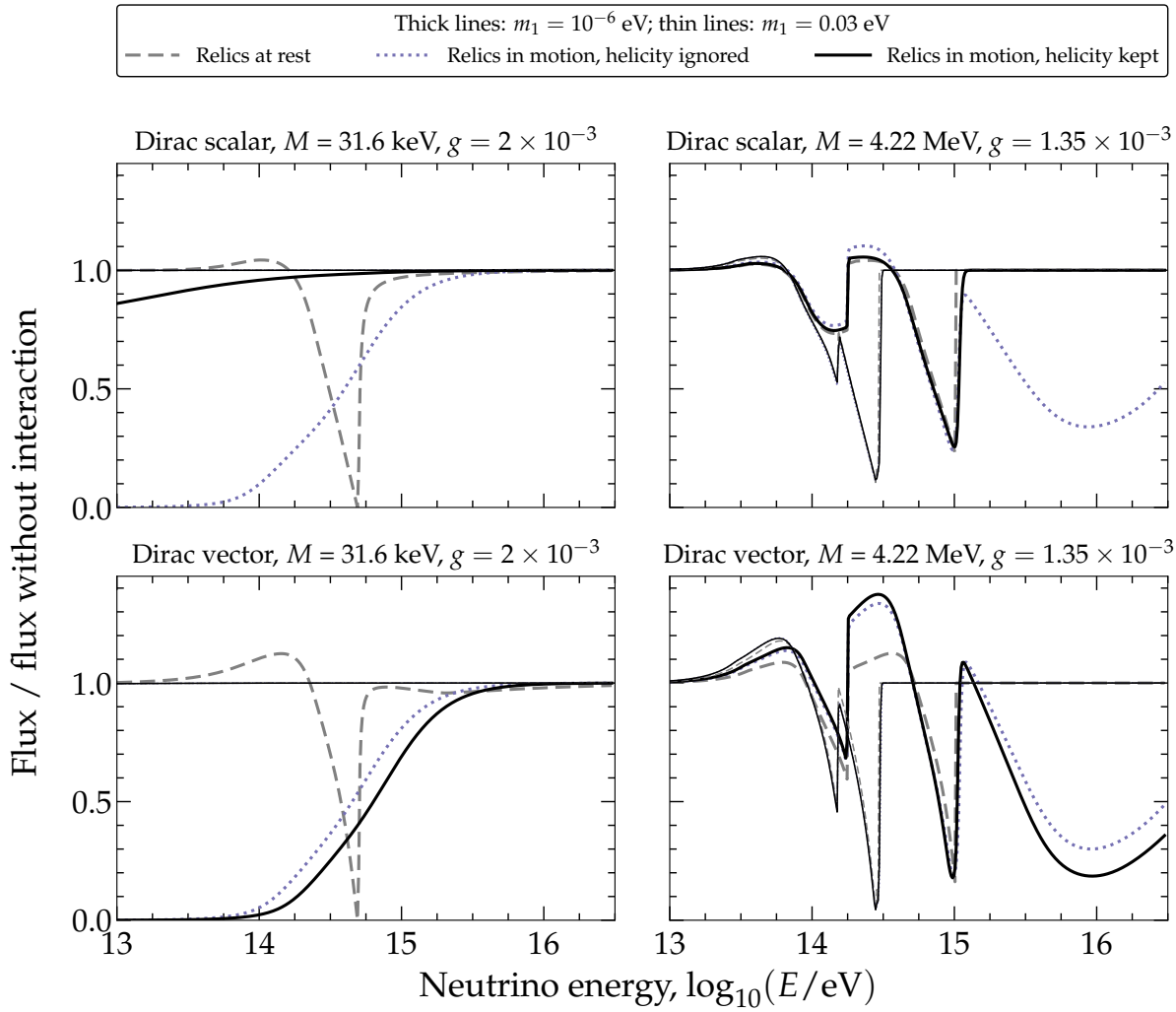


FIG. 12. ***Flux of high-energy neutrinos with secret interactions on Dirac relic neutrinos, with and without the helicity of the relics.*** The flux relative to that without a new interaction, for a mediator coupled to ν_e. The helicity matters most for a relativistic lightest relic: for the scalar, the resonance on it nearly disappears; for the vector, it deepens. See Appendix D for details.

are the effective squared masses $\mu_a(\sigma)$. A parent in state a with energy E' yields a daughter in state b with energy xE'; parent and daughter are projected on their own eigenvectors, evaluated at σ and $x\sigma$, respectively. Energy conservation fixes the transverse momentum of the daughter,

$$k_T^2 = (1-x)\,[x\,\mu_a(\sigma) - \mu_b(x\sigma)] - xM^2 \geq 0\,, \qquad \text{(E2)}$$

and the rate is $d\Gamma_{a\to b}/dx = |\mathcal{A}_{a\to b}|^2/[16\pi E'(1+z)]$, with the amplitude

$$\mathcal{A}_{a\to b} = \sum_{ij} c^*_{jb}(x\sigma)\, c_{ia}(\sigma)\, \mathcal{V}_{ji}\,. \qquad \text{(E3)}$$

Here, $c_{ia} = \langle \nu_i | a \rangle$ and $\mathcal{V}_{ji}$ is the vacuum amplitude for ν_i to decay into ν_j and the mediator. The potential does not act as a mass in the amplitude; it enters only through the eigenvectors and the kinematic window. A state cannot decay into itself or into a state of the same kind that lies above it, so the lowest state of each kind decays only by flipping its helicity. A potential also opens the decay of a neutrino state into the antineutrino state with the same label, which is closed in vacuum. Without a potential, these rates reduce to those of Eq. (E1).

This treatment is valid as long as the width of every state stays well below its separation from the other states in effective mass squared, *i.e.*, as long as $\zeta_a = 2E'(1+z)\,\Gamma_a/\min_b |\mu_a - \mu_b| \ll 1$; the corrections are of order ζ_a. Near the line of perturbative unitarity, at g/M of 10 to 100 eV^{-1}, ζ_a reaches 0.1–0.5, so the decay rates there are uncertain by that fraction. [The outcome of the mapping of ζ_a in the final run.]

Two kinds of decay are left out. First, we take the mediator as invisible. Its decays back into neutrinos are open only for $M > 2m_1$; they would add neutrinos at the lowest energies of the spectrum. [Decision B: the

outcome of the check.] Second, we neglect decays from the Sun on. There, they change a state by more than 10^{-3} only where the flavor already follows the potential, at $\sigma/\Delta m^2_{21} \geq 840$ in the Sun and $\geq 3.4 \times 10^5$ in the Earth; the lowest states, as which the flux arrives, never decay there.

Appendix F: The secret-interaction transport

Without a potential, we follow the mass states directly: $F_i(E, z)$ counts the neutrinos per unit energy in ν_i. For Majorana neutrinos, one population holds both helicities of each mass state, since the interaction treats them alike; we split it between neutrinos and antineutrinos in the proportions of the sources. For Dirac neutrinos, we follow the visible and the sterile states of each mass state separately, six populations in all. The populations evolve as

$$\frac{\partial F_i}{\partial t} = \mathcal{S}_i - \Gamma_i F_i + \sum_j \int_E^\infty dE'\, K_{ij}(E, E')\, \Gamma_j(E')\, F_j(E') - \Gamma_i^{\text{dec}} F_i + \sum_j \int_E^\infty dE'\, \frac{d\Gamma_{j\to i}}{dE}(E' \to E)\, F_j(E')\,, \tag{F1}$$

where Γ_i is the total rate of collisions on the relics, and $K_{ij}(E, E')$ is the energy spectrum of the neutrinos in state i that a collision of ν_j with energy E' produces. The last two terms are the decays of the neutrinos into the mediator (Appendix E). The rate Γ_i sums the s, t, and u channels, their interferences, and pair production [Fig. 3(b)–(e)], each averaged over the momenta of the relics (Appendix D). Each collision between two neutrinos produces two neutrinos. Each pair production produces four when the mediators decay into neutrinos, which they do at once. The products of the t and u channels at small momentum transfer return close to the incident energy; we treat their return as a drift in energy rather than as a transfer between energy bins.

For the scalar, we take the cross sections of Ref. [22], Appendix A, with two corrections. First, the Majorana mass states couple through the complex matrix $\lambda = U^T G U$ (Appendix A), which Ref. [22] takes as real; the two agree only without CP violation. Second, pair production sums the exchanged mass states coherently, which changes its cross section by factors of 0.08 to 2 with respect to the printed form. For the vector, we derive the cross sections from Eq. (4). We checked every cross section against an explicit computation of the amplitudes with Dirac matrices and helicity spinors.

We use this transport for secret interactions at every mass, and for the dark portal and $L_e - L_\alpha$ at MeV masses, where their potential is negligible. Below 1 keV, it also carries the scatterings with large momentum transfer of the dark portal and $L_e - L_\alpha$, for couplings above 10^{-4}.

Appendix G: The Earth's chords

A neutrino that reaches the detector from zenith angle θ_z crosses a chord of the Earth or, from above, only the ice over the detector. Along each chord, we evolve the density matrix by Eq. (9) in steps. Each step takes the exact mean density of the Preliminary Reference Earth Model [30] over its length, computed layer by layer. When the range of the mediator approaches the thickness of a layer, we use instead the density smoothed by the Yukawa factor of Eq. (2), which relaxes across each boundary between layers over a length $1/M$. The phase accumulated by the potential then equals its integral along the chord, whatever the number of steps; we use 25. The same integral gives the Earth's potential at the detector and, where the range exceeds the radius of the Earth, outside it.

We average the chords over the sky with equal weight per $\cos\theta_z$, within each band of declination. At MeV masses, the potential accumulates a phase of 10^5 to 10^8 rad across the Earth, so the off-diagonal blocks of the density matrix oscillate rapidly from chord to chord. An equal-weight mean over a finite set of chords would keep these oscillations as noise. Instead, we integrate each block against its phase over $\cos\theta_z$, which reproduces the exact average over the sky to 10^{-5}.

Appendix H: Existing bounds

[The bounds shown in gray, their sources and how they were obtained.]

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